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Selecting textual analysis tools to classify sustainability information in corporate reporting

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ABSTRACT

Information on firms' sustainability often partly resides in unstructured data published, for instance, in annual reports, news, and transcripts of earnings calls. In recent years, researchers and practitioners have started to extract information from these data sources using a broad range of natural language processing (NLP) methods. While there is much to be gained from these endeavors, studies that employ these methods rarely reflect upon the validity and quality of the chosen method—that is, how adequately NLP captures the sustainability information from text. This practice is problematic, as different NLP techniques lead to different results regarding the extraction of information. Hence, the choice of method may affect the outcome of the application and thus the inferences that users draw from their results. In this study, we examine how different types of NLP methods influence the validity and quality of extracted information. In particular, we compare four primary methods, namely (1) dictionary-based techniques, (2) topic modeling approaches, (3) word embeddings, and (4) large language models such as BERT and ChatGPT, and evaluate them on 75,000 manually labeled sentences from 10-K annual reports that serve as the ground truth. Our results show that dictionaries have a large variation in quality, topic models outperform other approaches that do not rely on large language models, and large language models show the strongest performance. In large language models, individual fine-tuning remains crucial. One-shot approaches (i.e., ChatGPT) have lately surpassed earlier approaches when using well-designed prompts and the most recent models.

1. Introduction

Scholars and practitioners from various domains such as management, finance, and accounting increasingly employ natural language processing (NLP) to extract structured and comprehensive information from unstructured textual data [1–5]. NLP techniques are particularly useful for quantitative-empirical research or in practical applications when structured quantitative data is not readily available. This issue is particularly relevant when searching for information about firms' sustainability [6]. Capturing firm-level sustainability reliably and comprehensively has become increasingly important for addressing global challenges such as the aggravated climate crisis, loss of biodiversity, and the spread of diseases [7].

One way to collect sustainability information from large corpora of text, including annual reports, firm news, and earnings calls [8–10], lies in using NLP methods. While many of these methods still rely on creating and using custom-made dictionaries [11–13], the emergence of more advanced techniques rooted in machine learning has spurred the development of a wide variety of techniques for NLP ranging from topic modeling to large language models (LLMs) [1,14–16]. These methods have gained popularity in multiple domains [10,14,16–19] and have been used to extract information on such topics as innovation [1,20], corporate culture [16], digitalization [9], emotions [21,22], and sustainability [23].

Our review of the literature on using various NLP methods to process information on constructs in text yields three important observations. First, we note that users rarely reflect upon the choice of NLP method. Second, users often do not comprehensively validate their method or report how well the tools they employ capture the constructs of interest in the texts. This lack of validation is particularly problematic where context-sensitive language is present. For instance, the terms “regulatory environment” and “environmental regulation” have very different

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meanings in the context of sustainability [24]. Finally, we observe that while the extraction of sustainability information based on NLP is of increasing interest to many users [25,26], many use cases tend to rely on dictionary approaches and topic models [1,11,13], neglecting more advanced NLP methods, including pre-large language models (pre-LLMs), such as word embeddings, LLMs, such as bidirectional encoder representations from transformers (BERT), and generative pre-trained transformers (GPT). In this study, we seek to address these challenges by providing a comparison of the model validity and quality of different NLP methods for extracting sustainability information.

To evaluate the utility of different methods for extracting sustainability information, we collected 75,000 sentences from the business description and management discussion sections of 10-K annual reports. Three researchers coded each sentence as either containing sustainability information or not. Using this manually coded data as ground truth labels, we then compared fifteen tools with different levels of complexity from four common textual analysis approaches to evaluate their performance in replacing manual information extraction.

Specifically, we compared (1) three dictionary approaches, (2) two topic models (one supervised and one unsupervised), (3) word embeddings, and (4) nine LLM approaches, including six BERT-based models and three different prompting approaches using ChatGPT. We took the three dictionaries from studies designed to identify sustainability information in texts [23,27,28]. To create a level playing field of training data for the other NLP methods, we used Wikipedia articles that were labeled as related and unrelated to sustainability based on link relationships. We trained an unsupervised biterm topic model (BTM), a supervised latent Dirichlet allocation (sLDA), word embeddings based on word2vec, and a BERT model (SustainBERT) that is original to this study to identify sustainability information. Moreover, we used a BERTopic model and sentence similarities based on Sentence-BERT. In addition, we also compared two publicly available fine-tuned transformer models (ClimateBERT and FinBERT ESG). The BERT models were specifically fine-tuned to identify sustainability information. With the inclusion of ChatGPT, the study also introduced one-shot learning competitors. For ChatGPT, we included results for two models, ChatGPT 3.5 Turbo and ChatGPT 4, and two different prompts.

Our analyses yielded the following findings. First, topic models generally perform better than dictionaries and other pre-LLM approaches. Second, there is substantial variation in the quality of dictionary approaches, with some being surprisingly good but others showing a less convincing performance. However, none of these conventional approaches can compete with LLMs. Finally, ChatGPT (with comprehensive information provided in the prompt and in particular with ChatGPT 4) has recently surpassed previous approaches to using LLMs. Within the class of pre-generative artificial intelligence LLMs, well-trained models designed for a specific task (e.g., SustainBERT) provide a best-in-class performance.

This study contributes to the literature in at least four important ways. First, we validate the current popular practice of using topic models to extract structured and comprehensive information from unstructured textual data and find them to have a decent performance compared to other pre-LLM approaches. While dictionaries remain popular in many applications, we highlight how important the choice of dictionaries is. When the quality of a dictionary is unclear, it is advisable to move to more advanced approaches. Second, if possible, applications should generally employ LLMs for their superior performance. Third, our study further highlights that even in this class of methods, the choice of model remains crucial to achieving the best performance for the envisioned application. In the class of pre-generative artificial intelligence methods, our results indicate that users should rely on well-tailored, fine-tuned transformer models rather than general models.[1] Fourth, in the class of generative artificial intelligence methods, well-designed prompts and more advanced models recently surpassed earlier available approaches.

In Chapter 2, we provide an overview of the literature on sustainability and the different NLP methods that we examine in this study: (1) dictionary-based approaches, (2) topic modeling, (3) word embeddings, and (4) LLMs (including generative artificial intelligence). Subsequently, we describe our data collection and design of the benchmark in Chapter 3, followed by the implementation of NLP methods in Chapter 4. In Chapter 5, we show the results of our comparison. Finally, we discuss limitations and future research in Chapter 6, followed by the conclusion in Chapter 7.

2. Literature review

2.1. An overview of sustainability-related research

Humanity is confronted with grand challenges such as climate change, loss of biodiversity, and social issues such as poverty and famine [29]. Tackling these grand challenges depends on a transition towards sustainability which can be defined as acting in a way that “meets the needs of the present without compromising the ability of future generations to meet their own needs” [30]. Yet, this transformation can only be successful if firms operate more sustainably. In a corporate context, this could include among others reducing emissions, responsible resource use, protecting biodiversity, improving working conditions, ensuring the safety of products and services, and good corporate governance. More broadly, sustainability at a firm level can be divided into an ecological, social, and economic dimension. Understanding sustainability at a firm level is an important endeavor in research from different domains such as management [7,29], finance [31], and information systems [32].

Given its multi-disciplinary nature, research on sustainability combines a variety of different research methods and philosophies from conceptual, qualitative, and quantitative spheres. Even though prior research yielded important insights into understanding sustainability, advancing empirical sustainability research is limited by the data available to researchers. Especially, quantitative research on the sustainability of firms is affected by data limitations for three main reasons. First, many aspects of sustainability are not consistently reported (e.g., CO2 emissions) which hinders comparisons and quantitative analyses [7,31]. Second, some aspects are hard to measure by their nature. For instance, extracting sustainability targets from text can be challenging due to the large amount of text. Finally, the broad definition of sustainability makes it hard for researchers to measure this abstract construct at an aggregated firm level [7,31]. Therefore, comprehensive ways to automatically process sustainability information could help to advance future research and facilitate decision-making in practice.

Recent studies employed NLP to extract sustainability information from corporate texts. For instance, Vaupel et al. [23] use a dictionary to measure firms' orientation towards sustainability and analyze its effect on sustainability performance. Bingler et al. [26] study corporate sustainability disclosures to assess climate-related risks using a BERT model. Szekely & vom Brocke [33] investigate the information content of firms' sustainability reports to derive propositions for research and practice. However, studies rarely reflect on their method choice or validate their method against other NLP methods. This can result in biased results as sustainability information is highly context-sensitive.

[1] Our study particularly addresses the situation where researchers do not aim to label training data. Based on our later results in the robustness section, we note that performance could be further optimized by making the investment and label data for a specific problem.

For instance, "regulatory environment" and "environmental regulation" have very different meanings [24]. Therefore, the question arises which NLP tools are most suitable for extracting sustainability information from corporate reporting.

2.2. An overview of textual analyses-related research

Dictionary-based approaches assess the extent to which a given text reflects the vocabulary of a dictionary. For instance, one can use the relative number of words in a text (often weighted by the inverse document frequency) that are also contained in a dictionary to measure the construct that the dictionary aims to identify. While dictionaries have often been applied to measure sentiment [34], they can also be used to measure abstract constructs such as sustainability [23] and corporate culture [16]. Even though the dictionary approach is easy to apply and is still used in various applications, it comes with the drawback of requiring a manually determined set of words that are characteristic of the construct of interest. If such a dictionary is not readily available, users have to develop and validate a dictionary of their own, which is a difficult and time-consuming task. Furthermore, the results of such an approach always contain some discretionary judgment on the list of words, making them subject to criticism [35]. Moreover, as the number of words is usually limited, the approach easily misses important information in a text because some words might not be included in the dictionary. Finally, the dictionary approach is not sensitive to semantic relationships because it only counts words. This drawback can lead to interpreting words outside of their context which is especially relevant for sustainability [24].

Topic modeling is an alternative method that circumvents the need to manually develop a dictionary [1,14]. Topic modeling relies on the assumption that texts comprise a given set of topics, and those topics determine the respective word distribution. Topic modeling therefore uses algorithms to fit word attributions and derive topics from an input of text [36,37]. These topics can be used to categorize text and determine their respective probabilities. Besides using topic probabilities directly, topic modeling can also generate dictionaries based on keywords that characterize topics. Prominent examples of topic modeling are latent Dirichlet allocation (LDA) [36] and bi-term topic modeling (BTM), which is particularly suitable for use with short texts [38]. While topic models require more technical effort than dictionary-based approaches, they are less subject to missing relevant words, and they make it possible to include some form of semantic relations as they are based on the distribution of words in a text. The choice of which topics to use in a study remains at the researchers' discretion. To limit this potential for bias, researchers could use a supervised version of the LDA, known as sLDA. However, sLDA can be used only when labeled data is available for training [39].

Word embeddings are models that use techniques related to deep learning [40] to convert each word of a vocabulary (or possible n-term phrases) into a multi-dimensional vector. These vector representations of words can then be used to represent the semantic meaning of words based on their position in the multi-dimensional vector space. If a researcher aims to determine if a text is related to a specific domain, they can determine the mean proximity of the words in the text to their position in the multi-dimensional vector space where the construct of interest is positioned. A researcher could also use word embeddings to generate dictionaries based on words in the proximity of a certain construct [16]. Further, it is possible to gauge the proximity between given texts and reference texts. Prominent algorithms to train word embedding models are based on word2vec, which uses deep learning to predict words following or surrounding a given word in a text [41], or GloVe, which relies on word correlations [42]. As is the case for topic models, word embeddings require some technical effort to train and apply. Word embeddings as a language model have been shown to contain valuable semantic information [41,43]. They can easily classify text after a researcher determines the seed words that mark the area of the vector space that is characteristic of a certain construct (e.g., sustainability).

Further developments in NLP have spurred the use of transformer-based textual analysis methods. After the advent of deep learning in the so-called ImageNet moment of 2012 [44], deep learning developed into the standard approach for the classification and processing of images and text. Textual analysis using deep learning has been frequently conducted through architectures such as convolutional neural networks or recurrent neural networks (with many additional developments such as long short-term memory networks). Recurrent neural networks evaluate individual words based on the words immediately before (and sometimes after) them. Transformer methods have developed more efficient ways to train models that evaluate words based on other words in the text and train particularly large neural networks—containing a hundred or even several hundred million or more trained parameters—on very large corpora of text [45]. In the case of BERT, for example, which paved the way for transformers, the model was trained on the BookCorpus and a complete copy of Wikipedia. Transformer models are trained in two different steps. In the first step, which is the computationally critical part, the procedure learns a language model by using text in which individual words or sentences are masked, and the model needs to recover them. From this step, the BERT model obtains some semantic understanding. These pre-trained models are then distributed; however, they are not yet fine-tuned to fulfill more specific tasks. Thus, the model needs to be trained in a fine-tuning step. In this second step, a BERT model can be adapted to determine certain constructs such as sustainability or tone in a text. Although BERT requires more effort to train than earlier approaches to measuring constructs in texts, it has the advantage of having a much deeper semantic understanding of the meaning of words in their context. LLMs, including BERT, have set new standards in performance in many NLP tasks [45,46] and in this way have revolutionized NLP.

Another application of transformers is generative artificial intelligence, with GPT being a prominent example. Generative models are not fine-tuned for some classification tasks but are used to generate text based on user inputs [47]. This could range from simple prompt completion to full conversational agents. A well-known application of generative artificial intelligence is the online chatbot ChatGPT, which performs tasks for users such as answering questions or writing (small) texts. ChatGPT set a new standard regarding the usability of LLMs, as users can operate it via a chat interface without needing any coding skills. This ease of use paired with its advanced capabilities allowed the application to reach 100 million users within just three months. What is a remarkable distinction between earlier methods and generative artificial intelligence is that generative artificial intelligence can be immediately used for a wide range of language-based tasks without necessarily being trained to fulfill any specific task. Generative artificial intelligence such as ChatGPT can therefore be used to classify sustainability information without prior fine-tuning (i.e., in a one-shot approach). Thus, the application of generative artificial intelligence could often be more easily introduced compared to earlier models due to low (local) technical requirements and well-defined interfaces. In the case of ChatGPT, the service is available via an API offered by OpenAI. Table 1 provides an overview of the reviewed NLP methods.

3. Data collection and the design of the benchmark study

3.1. Textual data

In this study, we used two types of textual information. First, we used textual data to be classified using NLP (testing data). Second, we needed textual data to train the NLP models (training data). This reflects the situation that data used in applications such as corporate reporting or news will usually not come with labels. Still, data for training should contain readily available labels. Users will naturally prefer solutions that require low effort for manual labeling before actually fitting models.

Table 1
Overview of textual analysis methods.

Method	Type	Logic	Approach	References
Dictionary-Based Approach	Pre-LLM	A dictionary comprises words that stand for a concept to measure. Then, the relative amount of words from the dictionary in a given text is used to measure the construct.	- Develop/choose a dictionary - Calculate relative word count (possibly weighted by inverse document frequency)	[34]
Unsupervised Topic Modeling	Pre-LLM	A topic model identifies thematic compositions in a given set of unlabeled texts based on word distributions. The model can be used to calculate topic probabilities that measure a construct for a new set of texts.	- Choose a topic modeling algorithm - Set number of topics and fit the model - Choose topic - Calculate topic probabilities	[36,38]
Supervised Topic Modeling	Pre-LLM	Supervised topic models abstract topics from text based on labels and word distributions. The model can be used to calculate topic probabilities that measure a construct for a new set of texts.	- Prerequisite: labeled text - Choose a topic modeling algorithm - Fit the model - Calculate topic probabilities	[39]
Word Embeddings	Pre-LLM	Word embeddings process words or texts by modeling their semantic meaning through a multi-dimensional vector space, which can be used to estimate proximities with constructs to measure.	- Choose the algorithm - Fit the model - Locate the construct of interest - Calculate proximities	[40,41]
Transformers (BERT)	LLM	BERT is a large pre-trained NLP model capturing contextualized text representations which can be fine-tuned with labeled text to estimate probability scores for a given construct.	- Prerequisite: labeled text - Choose a fine-tuned model or individually fine-tune a pre-trained BERT model - Calculate probabilities	[45]
ChatGPT	LLM	ChatGPT is a generative large language model that can answer prompts, understand, and produce texts.	- Define prompt to request an assessment of entanglement with a construct - Run prompt via the ChatGPT API	[47]

Typical types of textual information are news, company announcements, social media, and annual reports [14,16,34,48]. For this study, we collected 10-K annual reports from the United States Securities and Exchange Commission (SEC) for all firms that were on the S&P 1500 between 2010 and 2020. We extracted the business description (Item 1) and the management discussion section (Item 7) from the reports, which are established data sources for textual analysis applications [49].

To enable an effective comparison among the approaches and to prepare the texts for manual coding, we divided the texts into individual sentences as the sentence is a meaningful unit that can easily be understood and evaluated by a human reader. From the 10-K filings, we drew a random sample of 75,000 sentences: 25,000 from the business descriptions, 25,000 from the risk factors sections of the business descriptions, and 25,000 from the management discussions. These 75,000 sentences formed the basis for our method comparison. To quantify sustainability using the textual analysis models, the sample of sentences was then further preprocessed. First, the text was converted into lowercase for all methods except ChatGPT. For BERT, the sentences were used in this form. For the other methods, numbers, punctuation, and special characters were removed. Additionally, we performed lemmatization. Finally, common English stop words were removed. This sample of cleaned text then served as the basis for the method performance comparison. The prompt for ChatGPT was created using the initial sentences without a transformation to lowercase letters.

3.2. Ground truth labels for the comparison of methods

To establish ground truth labels for our method comparison, the 75,000 sentences were manually triple-coded, with each sentence being assessed by three researchers. A sentence was coded as either containing information related to sustainability (=1) or not (=0). The coders were trained to code sustainability based on a predefined conceptualization of sustainability derived from the literature. We calculated Cohen's kappa (0.61) to assess the interrater agreement of our three coders [50] which suggests a substantive interrater agreement, so we can rely on the ratings as ground truth labels [51].

For our model comparison, we classified a sentence as related to sustainability if at least two coders identified sustainability-related information. This verdict of the manual triple-coding was used as the ground truth label in our benchmark comparison. In total 3.24% of the sentences were labeled as related to sustainability. We then later assessed the quality of classifications compared to the manual classification. To assess the quality of the proposed textual analysis approaches, we compared the models' correlations, precision, recall, and F1 scores. We also constructed receiver operating characteristics (ROC) curves and precision-recall curves (PRCs) and compared the respective areas under the curves (AUC). Moreover, we analyzed lift curves. Reviewing the correlations offered insights into the extent to which the different methods yield similar results and agree with the manual labels. ROC curve and PRC analyses allowed us to evaluate the discriminatory power of our measures by visually inspecting the shape of the curves and comparing the AUC [52,53].

3.3. Training of the NLP models

To develop models, one needs to consider that labeled data from corporate reporting and news is often unavailable and will only be available through extensive manual labeling. Therefore, we aimed to train models on labeled text data that is readily available even when users (either researchers or practitioners) are not willing to conduct this manual labeling. To achieve this, we constructed labeled data from Wikipedia. We determined the labels by making use of the links among Wikipedia articles. To collect texts that were labeled as being related to sustainability, we considered the Wikipedia article on sustainability as the starting point of our data collection, as this article provided an overview of relevant aspects of sustainability. To achieve a holistic and comprehensive set of sustainability texts, we collected all articles that were linked from the main article on sustainability. Additionally, we screened the set of resulting articles for further linked articles and added them to our list of articles. Finally, we downloaded and revised this list, deleting articles that were not related to sustainability (e.g., broad articles on countries or historical events).

As the 10-K annual reports were mostly unrelated to sustainability, the models needed to be trained to understand and identify the differences between sustainability language and other content. Hence, we needed text that was unrelated to sustainability and more related to general reporting. We collected texts on accounting and management, as these are likely typical topics in corporate reporting. We again used Wikipedia articles as the source, and we collected texts following the same procedure used with the article on sustainability. This procedure yielded three sets of articles: (1) on sustainability (1879 articles), (2) on accounting (3548 articles), and (3) on management (7983 articles). Finally, we applied the same preprocessing procedure that was used for the 10-K annual reports. All competing models were trained on the same set of Wikipedia articles except for the dictionary approaches, which rely on pre-defined dictionaries, and the publicly available, fine-tuned transformers that rely on existing fine-tuned models. ChatGPT was used by sending prompts to the ChatGPT API. Fig. 1 provides an overview of our research design. More precisely, it presents our approach to preparing the training and testing data. Moreover, we show how we employed and compared the different NLP methods.

4. Implementation of the NLP methods

4.1. Dictionaries

The first approach to measuring sustainability that we investigated in our study was sustainability dictionaries. The dictionary approach is well established in research from different domains, but it is dependent on the availability of dictionaries. To obtain existing dictionaries to measure sustainability, we reviewed the literature and selected three of them.[2] Pencle and Mălăescu [28] established a multi-dimensional dictionary (hereafter referred to as PM16) that consists of 1419 words and is closely related to the concept of corporate social responsibility which focuses more on the social dimension of sustainability. Baier et al. [27] developed a sustainability dictionary (hereafter referred to as B20) based on the environmental, social, and governance (ESG) framework. The dictionary comprises a list of 491 keywords across the three topics of environment, society, and governance that relate to 10 categories and 34 sub-categories. Therefore, it contains a broad definition of sustainability. Finally, Vaupel et al. [23] created a sustainability dictionary (hereafter referred to as V22) with a focus on firms' sustainability orientation (268 words) which is mostly aligned with the conceptualization of sustainability in our training data.

As the three dictionaries differ in their creation processes, goals, and number of keywords, we treated each as an individual method for our comparison. Therefore, we used each of the three dictionaries individually to calculate the relative word count of sustainability-related words in the sentences extracted from the 10-K annual reports. We presumed that the higher the relative word count, the more information on sustainability is present in a sentence.

4.2. Topic modeling

4.2.1. Unsupervised topic modeling

As the sentences that are classified in our benchmark study are relatively short compared to full-text corpora, we use biterm topic models (BTMs), an unsupervised topic modeling approach that performs more favorably with short texts than other popular topic modeling techniques such as LDA [38]. This is because BTMs rely on word co-occurrences in a local context instead of overall word occurrences [38].

A BTM relies on a specified number of topics to generate the model. Therefore, we calculated BTM models with three to 30 topics in intervals of one and evaluated each model based on its Akaike information criterion (AIC), which measures the log-likelihood of the model while penalizing a high number of topics (i). We selected a model with 30 topics, as it showed the lowest AIC, which indicated that this model best fit the data.

$$AIC = 2 \times i - 2 \times log(likelihood)$$

To measure sustainability in the sentences that we extracted from the 10-K annual reports, we needed to identify which of the 30 topics from our topic model best described sustainability. To do so, we reviewed the correlations of each topic with the sustainability labels. Following Bellstam et al. [1], we chose the topic that had the highest correlation with the Wikipedia categories. The correlation of the topic with the sustainability label was above 0.5. Finally, we calculated the topic probability of the topic that we identified as closest to the sustainability labels for the 10 K filing sentences. This topic probability served as the BTM classifier in our comparison.

4.2.2. Supervised topic modeling

The second topic modeling approach in the comparison was supervised. We decided to use an sLDA to compare it with the unsupervised BTM. Supervised topic modeling does not require the researcher to specify the number of topics, but it requires training on data that includes labels [39]. In our case, we used the affiliation of each Wikipedia article to one of the two main themes—either sustainability on the one hand or management and accounting on the other—as the label. This allows the supervised topic modeling algorithm to identify texts (i.e., Wikipedia articles) that are related to sustainability. We estimated the sLDA model, which we then used to calculate the topic probabilities for the 10-K sentences. As the topic of sustainability resulted from the classification of the Wikipedia articles, there was no need to calculate correlations to determine the sustainability topic in this case. Therefore, only the probability of the sustainability topic on the 10-K filing sentences was compared to the benchmark.

4.3. Word embeddings

As an alternative approach to topic modeling, we generated a word embedding model based on the work of Mikolov et al. [40]. To do so, we first trained a word2vec model. The word2vec model transformed the words of our input texts into multi-dimensional vectors that represent a word's meaning. This allowed us to quantitatively assess the semantic similarity of vectors (i.e., words or texts). In the vector space, vectors that are close to each other are assumed to have a similar meaning. As we were interested in measuring sustainability, we then calculated the word vector of the term “sustainability” in our word2vec model. Furthermore, we transformed each word in the sentences extracted from the 10-K annual reports into a vector. The measure for sustainability then reflected the cosine distance between the mean vector of a sentence and the vector of the word “sustainability.”

4.4. Large language models

4.4.1. Pre-generative artificial intelligence approaches

Transformers are a deep-learning neural network architecture for processing text. Devlin et al. [45] established BERT and demonstrated the superior performance of transformers for several important NLP tasks. BERT is trained in an unsupervised way, without a specific later downstream use case, on large corpora of text [45]. This is done using left-out word prediction and sentence prediction tasks. BERT is then later fine-tuned to perform downstream tasks such as classification.

We first downloaded the English *BERT-base-uncased* model for further processing. The critical step was to fine-tune SustainBERT for identifying phrases related to sustainability, which requires labeled data. For most constructs, no labeled data is readily available. This is likely to be one of the critical hurdles that often hinder the adoption of BERT in textual analysis of corporate reporting. Bingler et al. [26] used

[2] We acknowledge at this point that the dictionaries could possibly follow a different definition of sustainability compared to our training data.

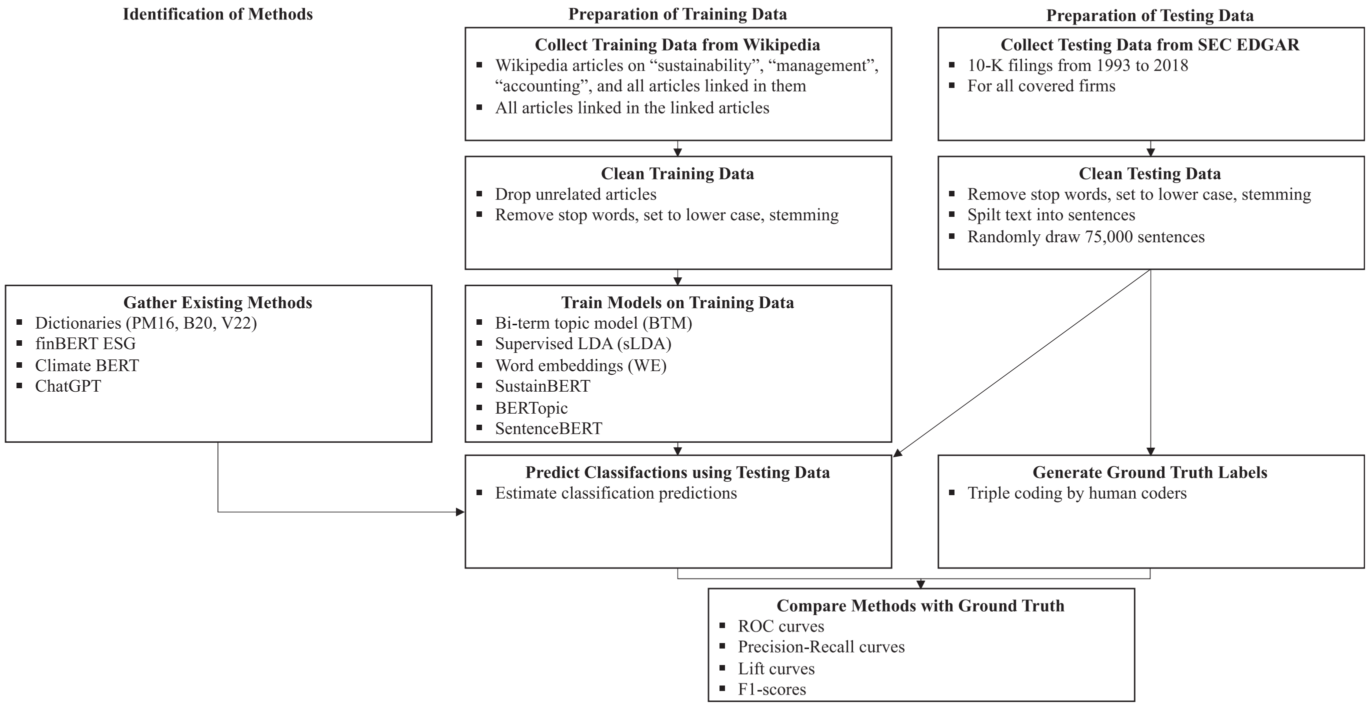

Fig. 1. Research design process.

manual coding to produce labeled data for such purposes. However, for many textual analysis settings, manually coding large parts of a corpus requires a lot of effort. To circumvent this time-consuming coding, we used the Wikipedia categories to derive labeled data. We trained SustainBERT to distinguish between sustainability-related articles on the one hand and management- or accounting-related articles on the other. Later, we used the SustainBERT model to classify how likely it was that a benchmark sentence from the 10-K filings was related to sustainability or not. The fine-tuned transformer models that are available online, FinBERT and ClimateBert, were retrieved from the Hugging Face repository. In applying these models, the fine-tuning step was not necessary, and they were used to classify the 10 K filing sentences directly. For Sentence-BERT, we downloaded the model *all-MiniLM-L6-v2*, split the Wikipedia article on sustainability in sentences, and calculated the mean cosinus similarity to the 10 K filing sentences [54]. For BERTopic, we used a similar procedure as used in the topic models [55]. We trained the topic model on the set of Wikipedia articles, determined the topic that had the strongest correlation with the sustainability category, and then used its prevalence in the 10 K filings as the score. We then also built a model by splitting the Wikipedia articles into sentences and cleaning the sustainability article sentences using the similarities to the sustainability article using Sentence-BERT. The cleaned sentences are used to fit a BERT classifier.[3] The resulting model will be labeled Sentence-SustainBERT in the following.

There are existing transformer models fine-tuned to measure sustainability. These models differ from each other and the SustainBERT model that we fine-tuned for the present study. While existing transformer models for sustainability are fine-tuned using manually labeled data, the approach that we chose for this paper has the advantage of not relying on the manual coding of sentences, which significantly reduces the effort to create the model and the subjectivity. Therefore, we were interested in comparing the classification quality of existing BERT models with the SustainBERT approach. Reviewing the sparse literature related to BERT models that measure sustainability, we found two suitable models: ClimateBERT and FinBERT ESG.

First, the ClimateBERT model developed by Bingler et al. [26] was designed to identify and measure climate-related information in corporate disclosure which diverges from our broader definition of sustainability.[4] It was trained using 17,300 manually labeled sentences from annual reports that cover climate-related information. While this approach focuses on only one dimension of sustainability, it is worth comparing it as it is well-known in the research community. Second, Huang, Wang, and Yang [56] created FinBERT ESG and also published an adaption that was fine-tuned on 2000 manually labeled sentences to extract ESG-related information from corporate reporting. This broader approach is more in line with our definition of sustainability. In the following section, we compare the classification quality of the two BERT models with the SustainBERT model that we developed for the present study.

4.4.2. Generative artificial intelligence models

For generative artificial intelligence, we used a variation of two versions of ChatGPT, *ChatGPT 3.5 Turbo* and *ChatGPT 4*. We further created the following two prompts: (1) a simple prompt stating: “On a scale from 0 to 100, how entangled is the following statement with sustainability: ‘[10K filing sentence]’ Please state the number first and then one sentence as an explanation.” and (2) a more comprehensive prompt: “We need to classify sentences from firms' annual reports in being related or unrelated to sustainability for a research paper. Sustainability is defined as meeting the needs of the present without compromising the ability of future generations to meet their own needs. In a corporate context, this could include among others reducing emissions, responsible resource use, protecting biodiversity, improving working conditions, ensuring the safety of products and services, and good corporate governance. On a scale from 0 to 100, how entangled is

[3] We thank an anonymous reviewer for this suggestion.

[4] It should be noted that ClimateBERT is more focused on climate related aspects of sustainability and in this way follows a more narrow definition of sustainability. We still include it as a well-known comparison. FinBERT ESG covers ecological, social, and governance aspects, which is a more comparable comprehensive definition.

the following statement with sustainability: '[10K filing sentence]' Please state the number first and then one sentence as an explanation." From these prompts, the second reflects a more detailed description of the role and context as well as examples that made the task more detailed. The prompts were then sent to the ChatGPT-API and the resulting numerical assessments were extracted using regular expressions. This assessment then constituted the ChatGPT predictions in the comparison.

5. The classification quality of NLP methods

5.1. Comparison of the pre-large language model methods

To assess the classification quality of the different methods and their consistency, we first analyze precision-recall curves (PRC) and lift curves (LC). We then proceed to discuss a number of common performance metrics. PRCs plot the precision (ratio of true positives to the total predicted positives) and recall (ratio of true positives to the total actual positives) for varying levels of the threshold for positive classifications. LCs divide the sample into bins based on the predicted likelihood that a model assigns each case as being positive. The lift is calculated for each bin. It is defined as the ratio of the proportion of positive cases in the bin to the proportion of positive cases in the entire sample. This ratio indicates how much better the model is at identifying positive cases in the bin compared to random guessing.

A visual inspection of the curves indicates differences in the classification quality of the methods, which generally underlines the need for evaluation (see Fig. 2). Using SustainBERT as a reference, LLMs perform better than the pre-LLM methods. Only in the right part of the plot, which is rather uninformative for realistic settings, do the curves converge to a certain degree. The curve that corresponds to Dictionary V22 is well below SustainBERT but still significantly above all other curves until it reaches a tipping point at a recall of around 0.5, beyond which the curve falls steeply. This tipping point can be explained by the dictionary's number of words. SustainBERT shows the highest precision at most recall levels. This means that SustainBERT is identifying more positive cases (i.e., sustainability-related sentences) at a higher precision than the other models, which indicates superior performance. The BTM follows closely after Dictionary V22. One can generally observe a significant variation in the performance of dictionaries by comparing the curve for Dictionary V22 with the curves of Dictionary B20 and Dictionary PM16. The LCs are very consistent with SustainBERT displaying the highest lift, Dictionary V22, and the topic model following closely after, and some variation within the dictionary performances.

To quantify the performance, we calculated the area under the receiver operating characteristic curve (ROCAUC),[5] the area under the PROC (PRAUC), the precision, the recall, and the F1 score which serve as an indicator of the classification quality (see Table 2). Comparing the ROCAUC of our methods confirms that the SustainBERT model representing the large language models classifies the sentences most closely to the human coding as it has the highest ROCAUC at 0.84. The measure based on Dictionary B20 achieves a slightly lower score (ROCAUC = 0.82), followed by Dictionary PM16 (ROCAUC = 0.78) and the BTM (ROCAUC = 0.78). Word embeddings (ROCAUC = 0.74)[6] and Dictionary V22 (ROCAUC = 0.72) provide lower but still decent classification quality. The sLDA (ROCAUC = 0.62) shows the lowest values, which is a surprise, as one would expect supervised training to capture the construct of interest more accurately than unsupervised models.

Reviewing the PRAUC of the models confirms the visual interpretation of the curves. The SustainBERT model has the highest PRAUC (0.36), followed by Dictionary V22 (0.22) and the BTM (0.19). The other models perform substantially worse (see Table 2). Consequently, analyzing the PRC shows that SustainBERT performs best in classifying sustainability-related information in 10-K sentences. The well-calibrated dictionary and topic model perform worse than SustainBERT but still achieve much better results than the other approaches. Both ROC and PRC evaluate the models independent of classification thresholds. To make the differences between the models more tangible, we analyze the performance of the classifiers based on an exemplary threshold that allows us to calculate precision, recall, and F1 scores. In the following calculation, our exemplary threshold is set to the 95% quantile of each score. This would indicate that 5% of the sentences would be classified as related to sustainability (as noted above, the actual percentage of sentences labeled as sustainability-related is 3.24). Values above this threshold are classified as positive cases, i.e., related to sustainability. The results presented in Table 2 show that the measure based on SustainBERT performs best, followed by Dictionary V22. SustainBERT achieves the highest recall and precision rates and the highest F1 score. This means that at the exemplary threshold, the classifier based on SustainBERT can identify more (true) positive cases than any other model while being the most precise, i.e., with the fewest errors in positive predictions. Therefore, the analyses of our exemplary threshold support the notion that SustainBERT is the most favorable model as it identifies the most positive cases with the highest precision. While Dictionary V22 performs well, the three dictionaries yield very different results, which highlights the importance of choosing a well-calibrated dictionary for this approach. In sum, LLMs seem less likely to produce bias.

5.2. Comparison of pre-generative artificial intelligence large language models

While SustainBERT performs well in comparison to established textual analysis methods, we will now compare different specifications of LLMs using similar plots and metrics as in the previous analysis. A visual inspection of Fig. 3 shows that the PRCs of the LLMs tend to be on higher levels than the established textual analysis methods. For low levels of recall, FinBERT, ClimateBERT, Sentence-BERT, and Sentence-SustainBERT have particularly high levels of precision. FinBERT, ClimateBERT, and Sentence-BERT then have a significant drop in precision at much lower recall rates than the ones of SustainBERT and Sentence-SustainBERT. This indicates that the proposed SustainBERT as well as the Sentence-SustainBERT model can maintain higher precision at recall rates above around 30% while FinBERT ESG, ClimateBERT, and Sentence-BERT can achieve high precision at lower recall rates. Putting this finding more into context, the aforementioned were very sure about identifying about 20% of the actual sustainability sentences while being wrong when aiming to identify 50% of the sustainability sentences about four out of five times. The newly developed SustainBERT and the Sentence-SustainBERT identified 50% of the cases while being correct about half of the time. The specific weaknesses and benefits of these comparative performances are subject to the specific applications the BERT models are used in. However, the Sentence-SustainBERT excels in having a very favorable performance in both cases. The BERTopic model performs weaker than the other LLMs for low levels of recall but has an intermediate performance at a recall of around 50%. The lift curves are again in line with this assessment.

Further quantifying the results by calculating the area under the PRC (see Table 3) shows that leveraging BERT for topic modeling improves the performance of the topic model but still falls behind all other LLM approaches ($PRAUC_{BERTopic} = 0.22$). The public models ClimateBERT and FinBERT perform better ($PRAUC_{FinBERT\ ESG} = 0.28$, $PRAUC_{ClimateBERT} = 0.30$) and at similar levels as Sentence-BERT ($PRAUC_{Sentence\text{-}BERT} = 0.30$). SustainBERT and Sentence-SustainBERT outperform the aforementioned LLMs, with a PRAUC of 0.36 and 0.39. Finally, we

[5] Prior research suggests assessing precision and recall curves instead of the receiver operating characteristics curve when the data is heavily imbalanced [57].

[6] As a robustness check, we evaluated a doc2vec word embeddings model which performed slightly worse than the word2vec model.

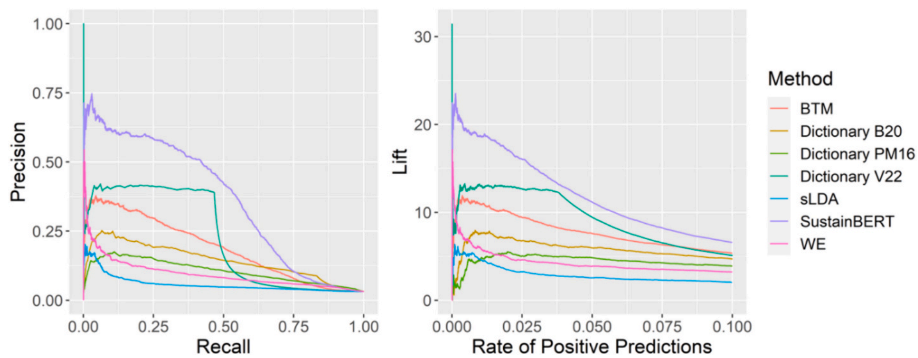

Fig. 2. Precision-recall and lift curves of pre-large language models.

Table 2
Performance metrics of pre-large language models.

Model	ROCAUC	PRAUC	Prec	Rec	F1
Dictionary PM16	0.78	0.10	0.16	0.24	0.19
Dictionary B20	0.82	0.14	0.19	0.28	0.23
Dictionary V22	0.72	0.22	0.30	0.48	0.37
BTM	0.78	0.19	0.24	0.38	0.30
sLDA	0.62	0.06	0.08	0.13	0.10
WE	0.74	0.09	0.12	0.20	0.15
SustainBERT	0.84	0.36	0.32	0.58	0.43

Note: BTM = Bi-Term Topic Model; sLDA = Supervised Latent Dirichlet Allocation; WE = Word Embeddings; BERT = Bidirectional Encoder Representations from Transformers.

compare precision, recall, and F1 scores of the LLMs at the exemplary threshold of the 0.95 quantile, as we did in the main analyses. Comparing the values confirms that SustainBERT and Sentence-SustainBERT also perform very favorably in precision, recall, and F1 score. The results show that these two approaches—fine-tuning BERT without manual coding and further enhancing it by sentence similarities derived from Sentence-BERT—achieve a classification quality that is at least equal to that of the other models, which were partly fine-tuned using significant effort in human labeling. Therefore, custom models can be fine-tuned with labels such as those from Wikipedia data to achieve state-of-the-art performance. This result points towards users ideally developing custom fine-tuned models for their applications within the class of pre-generative artificial intelligence LLMs.

5.3. Comparison of generative artificial intelligence large language models

In this chapter, we compare (1) ChatGPT 3.5 using a simple prompt and (2) an extended one, (3) ChatGPT 4 using the same extended prompt, and (4) SustainBERT. A visual inspection of the PRC (see Fig. 4) shows that the scores based on ChatGPT 3.5 with the extended prompt are similar to SustainBERT. The ChatGPT 3.5 with the more simple prompt is mostly below the curve of SustainBERT. The model based on ChatGPT 4 seems to outperform all other approaches. This notion is supported by the PRAUC values ($PRAUC_{ChatGPT\ 4} = 0.67$). SustainBERT ($PRAUC_{SustainBERT} = 0.36$) and ChatGPT 3.5 with the extended prompt ($PRAUC_{ChatGPT\ 3.5\ Ext.} = 0.35$) perform closely following and at similar levels vice-versa as the curves indicate (see Table 4). The uplift in performance from SustainBERT and ChatGPT 3.5 to ChatGPT 4 is very substantial. The exemplary F1 score, precision, and recall rates as well as the lift curves confirm the before-mentioned results. Consequently, our results show that, overall, ChatGPT can produce the best classification results. However, the performance of ChatGPT depends on the design of the prompt and especially on the version of ChatGPT that is used.[7]

6. Limitations and future research

This study does not come without limitations which open avenues for future research. First, while this study provides nuanced insights into how different NLP methods perform in classifying sustainability information in corporate reporting, these methods can be employed both in research and practice to support addressing questions related to a transformation towards a sustainable development. Therefore, future research can use the insights into NLP methods to leverage their potential to advance our understanding of sustainability and drive change for good. Furthermore, our study does not provide insights into whether and how a sub-optimal NLP method might influence the results of a study or a practical application. Future research can investigate this potential risk of method choice and research using earlier approaches for consistency.

Second, our method comparison relies on human judgment as the ground truth. As the interrater agreement could potentially bias our results, we employed several measures to address this issue. Hence, we calculated Cohen's kappa indicating a substantive interrater agreement with a value of 0.61. To check for robustness, we estimated all performance metrics based on the coding of each coder individually (available on request). This analysis supports our results yielding a similar ranking of methods as in our main analyses.

The design of the study then particularly addresses the situation when labeled text data for sustainability is not available. This is determined to best reflect the situation of researchers and practitioners using these methods. If researchers or practitioners are willing to make the effort and collect labeled data, there is potential for further performance increases in the self-trained models. We particularly conducted a robustness check by training a BERT classifier on parts of the labeled data and tested its performance on the remaining part (available on request). The performance exceeded that of ChatGPT 4 underlining the usefulness of well-tailored models.

Moreover, in this study, we only compare NLP methods in classifying sustainability which is highly relevant for both research and practice. However, the contributions are informative outside of the domain of sustainability. The capabilities of NLP can be utilized in a broad range of other use cases. Prior research showed examples of NLP being used to measure other complex constructs such as corporate innovation [1] or corporate culture [16]. Therefore, future research can also investigate

[7] In addition, we also conducted analysis steps using open source generative models, and Llama 2 (llama-2-13b-chat-hf) in particular. The performance was below the ChatGPT models and the pre-generative artificial intelligence large language models (ROCAUC of 0.71 and PRAUC of 0.09).

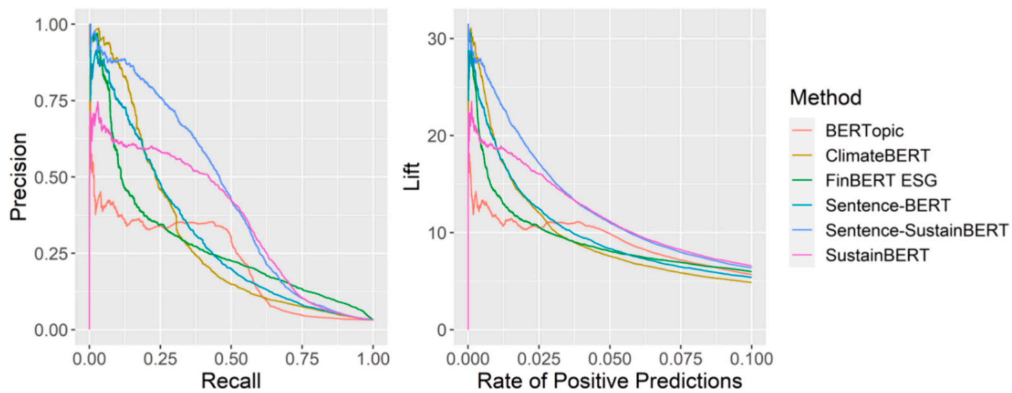

Fig. 3. Precision-recall and lift curves of the large language models.

Table 3
Performance metrics of different large language models.

Model	ROCAUC	PRAUC	Prec	Rec	F1
FinBERT ESG	0.88	0.28	0.26	0.40	0.31
ClimateBERT	0.80	0.30	0.24	0.38	0.29
BERTopic	0.75	0.22	0.31	0.49	0.38
Sentence-BERT	0.83	0.30	0.26	0.42	0.32
Sentence-SustainBERT	0.66	0.39	0.33	0.52	0.40
SustainBERT	0.84	0.36	0.32	0.58	0.43

Note: BERT = Bidirectional Encoder Representations from Transformers.

Table 4
Performance metrics of GPT models.

Model	ROCAUC	PRAUC	Prec	Rec	F1
ChatGPT 3.5 Simple Prompt	0.83	0.25	0.32	0.38	0.35
ChatGPT 3.5 Extended Prompt	0.86	0.35	0.32	0.51	0.40
ChatGPT 4 Extended Prompt	0.96	0.67	0.48	0.76	0.59
SustainBERT	0.84	0.36	0.32	0.58	0.43

possible differences in the performance of different NLP methods for other constructs and thereby improve our understanding of method choice for performance implications.

Finally, in a benchmark comparison, one could only compare a limited number of NLP methods. The study aims to include the most commonly used and thus relevant methods to this date. However, there are constantly newly developed approaches fueled by advancements in NLP and machine learning. Therefore, we assume that the landscape of available NLP tools will evolve over time. Consequently, future research could include new methods in this comparison to understand the potential benefits and drawbacks to inform method selection.

7. Conclusion

In this study, we (1) analyze the classification quality of different established NLP methods; (2) compare their performance with LLMs; and (3) compare a broad selection of LLMs including a custom fine-tuned transformer (SustainBERT), several other transformer approaches and multiple approaches to using ChatGPT in their ability to identify sustainability information. Our findings contribute to the literature in several ways. First, we evaluate the common practice of using topic models to extract information on the prevalence of certain constructs from text. Topic models have a performance that is favorable compared to alternative pre-LLM methods. These methods are therefore an appropriate choice when considering a lower complexity and making compromises regarding the performance. There was a large variation in the quality of dictionary approaches. While individual dictionaries deliver a surprisingly favorable performance for some settings, the performance is not consistent over dictionaries. Using dictionary approaches therefore introduces some level of uncertainty. Users should stick to well-validated dictionaries or rather turn towards more advanced methods. Second, LLMs deliver state-of-the-art performance and should generally be preferred for most use cases. Third, there is also some variation among different LLMs. The favorable performance of the custom fine-tuned large language model hints towards preferably fine-tuning transformers for the respective use case for most LLM approaches. Fourth, the most recent generative models (namely ChatGPT 4) have surpassed the performance of earlier LLM approaches and are in this way likely to make their way in many research settings using natural language processing.

CRediT authorship contribution statement

Frederik Maibaum: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources,

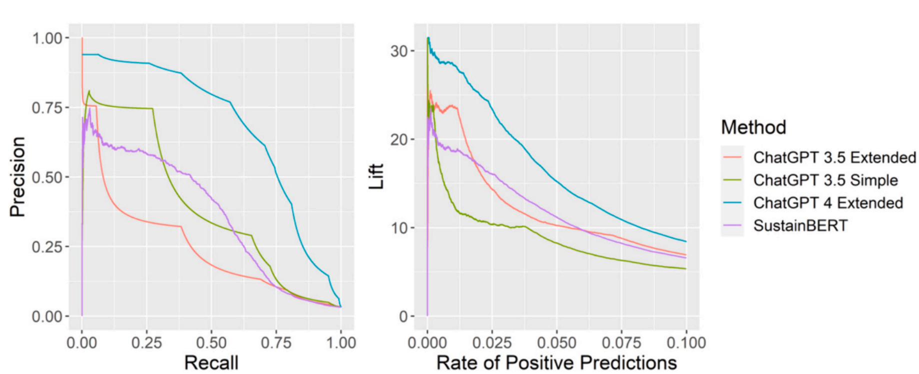

Fig. 4. Precision-recall and lift curves of GPT models.

Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Johannes Kriebel:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Johann Nils Foege:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on reasonable request.

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